Lab 7: Fuzzing for Input Validation Bugs

Step 2: Running the buggy student_grades.py

Works with the sample inputs, but crashes on an empty value in the marks CSV.

```
Lab 7 % python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG23CS123
Campus, Year and Dept: ('PES1', 23, 'CS')

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi kumar
Short Name: R. Kumar
Lab 7 % python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,,92
Traceback (most recent call last):
  File "student_grades.py", line 29, in <module>
    main()
    ~~~~^^
  File "student_grades.py", line 15, in main
    result = average_marks(user_input)
  File "student_grades.py", line 2, in average_marks
    marks = [int(m) for m in marks_csv.split(',')]
             ~~~^^^
ValueError: invalid literal for int() with base 10: ''
Lab 7 %
```

Step 4: Running test_student_grades.py on the buggy code

Hypothesis finds failing inputs for all three functions (e.g. `''` and None).

```
Lab 7 % python -m pytest --no-header test_student_grades.py
==================================== test session starts ====================================
collected 4 items

test_student_grades.py FFF.                                                          [100%]

========================================== FAILURES ==========================================
_____________________________ test_average_marks_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "test_student_grades.py", line 5, in test_average_marks_no_crash
  |     def test_average_marks_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 7, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |              ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 9, in test_average_marks_no_crash
    |     assert False, f"average_marks crashed with: {e}"
    | AssertionError: average_marks crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_average_marks_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         test_student_grades.py:8
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 7, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |                             ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 9, in test_average_marks_no_crash
    |     assert False, f"average_marks crashed with: {e}"
    | AssertionError: average_marks crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_average_marks_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         test_student_grades.py:8
    +------------------------------------
```

```
_____________________________________ test_parse_srn_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "test_student_grades.py", line 12, in test_parse_srn_no_crash
  |     def test_parse_srn_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 14, in test_parse_srn_no_crash
    |     parse_srn(s)
    |     ~~~~~~~~~^^^
    |   File "student_grades.py", line 6, in parse_srn
    |     return srn[:4], int(srn[6:8]), srn[8:10]
    |                     ~~~^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 16, in test_parse_srn_no_crash
    |     assert False, f"parse_srn crashed with: {e}"
    | AssertionError: parse_srn crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_srn_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 14, in test_parse_srn_no_crash
    |     parse_srn(s)
    |     ~~~~~~~~~^^^
    |   File "student_grades.py", line 6, in parse_srn
    |     return srn[:4], int(srn[6:8]), srn[8:10]
    |            ~~~^^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 16, in test_parse_srn_no_crash
    |     assert False, f"parse_srn crashed with: {e}"
    | AssertionError: parse_srn crashed with: 'NoneType' object is not subscriptable
    | assert False
    | Failing test case: test_parse_srn_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
_____________________________________ test_short_name_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "test_student_grades.py", line 19, in test_short_name_no_crash
  |     def test_short_name_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 21, in test_short_name_no_crash
    |     short_name(s)
    |     ~~~~~~~~~~^^^
    |   File "student_grades.py", line 9, in short_name
    |     first, last = full_name.split()
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 0)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 23, in test_short_name_no_crash
    |     assert False, f"short_name crashed with: {e}"
    | AssertionError: short_name crashed with: not enough values to unpack (expected 2, got 0)
    | assert False
    | Failing test case: test_short_name_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         test_student_grades.py:22
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 21, in test_short_name_no_crash
    |     short_name(s)
    |     ~~~~~~~~~~^^^
    |   File "student_grades.py", line 9, in short_name
    |     first, last = full_name.split()
    |                   ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "test_student_grades.py", line 23, in test_short_name_no_crash
    |     assert False, f"short_name crashed with: {e}"
    | AssertionError: short_name crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_short_name_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         test_student_grades.py:22
    +------------------------------------
=================================== short test summary info ====================================
FAILED test_student_grades.py::test_average_marks_no_crash - ExceptionGroup: Hypothesis found 2 d...
FAILED test_student_grades.py::test_parse_srn_no_crash - ExceptionGroup: Hypothesis found 2 disti...
FAILED test_student_grades.py::test_short_name_no_crash - ExceptionGroup: Hypothesis found 2 dist...
================================= 3 failed, 1 passed in 1.81s ==================================
Lab 7 %
```

Step 5: Running the fixed student_grades.py

Valid inputs give the same results; invalid inputs are reported instead of crashing.

```
Lab 7 % python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG23CS123
Campus, Year and Dept: ('PES1', 23, 'CS')

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi kumar
Short Name: R. Kumar
Lab 7 % python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,,92
Average Marks: Invalid input

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG2XCS12
Campus, Year and Dept: Invalid input

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi
Short Name: Invalid input
Lab 7 %
```

Step 6: Re-running test_student_grades.py on the fixed code

```
Lab 7 % python -m pytest --no-header test_student_grades.py
===================================== test session starts ======================================
collected 4 items

test_student_grades.py ....                                                            [100%]

====================================== 4 passed in 1.52s =======================================
Lab 7 %
```